CS-502 Fundamental Of Algorithms

Update MCQS For Quiz-4 File

Solve By Vu Topper RM

85% To 100% Marks



بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

For each vertex u ϵ (V − S) ,we associate a key _____.

A. key[v-s]
B. key[s]
C. key[u]
D. key[v]

In Kruskal's algorithm, the vertices will be stored in _____.

A. loops
B. sets
C. nodes
D. links

An edge (u, v) _____ E − A is safe if A _____{(u, v)} is viable.

A. ϵ,U
B. U,ϵ
C. ϵ,Π
D. Π,U

Which of the following statement is true.

A. Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is O(EV)
B. Kruskal algorithm is multiple source technique for finding MST.
C. Both I and II
D. Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges

Question No:1 (Marks:1) Vu-Topper RM

_____ Algorithm allows negative weights edges and no negative cost cycles.

A. Brute-force technique
B. Bellman-Ford Page 159
C. Dijkstra's
D. Print

Question No:2 (Marks:1) Vu-Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

In Prim's algorithm, we start with the _____ vertex r; it can be any vertex.

A. Root **Page 149**

B. Pivot

C. Leaf

D. Negative

Question No:3 **(Marks:1)** **Vu-Topper RM**

Overall Running time of Prim's algorithm is _____.

A. Θ(ElogE)

B. Θ(ElogV)

C. Θ((V+E)logV) **Page 152**

D. Θ((V+E)logE)

Question No:4 **(Marks:1)** **Vu-Topper RM**

Problems such as the shortest route between cities can be solved efficiently by modeling the road map as a_____.

A. Tree

B. Stack

C. Graph **Google**

D. Linked list

Question No:5 **(Marks:1)** **Vu-Topper RM**

In Kruskal's algorithm, the next edge is added to viable set A, if its adding does not induce a/an _____ .

A. Tree

B. Edge

C. Cycle **Page 147**

D. Vertex

Question No:6 **(Marks:1)** **Vu-Topper RM**

Bellman-Ford algorithm is used to solve _____ problems.

A. Flow of networking

B. All pair shortest path

C. Single source shortest path **Google**

D. Double source shortest path

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:7 **(Marks:1)** **Vu-Topper RM**

In Prim's algorithm, If the color of a vertex is _____,then it is in S otherwise not.

A. Blue
B. Gray
C. Black **Page 151**
D. White

Question No:8 **(Marks:1)** **Vu-Topper RM**

Keeping in mind the shortest-path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are_____and Edges are _____which may be wired or wireless.

A. Internet, routers
B. Routers, internet
C. Communication links, routers
D. Routers, communication links **Page 153**

Question No:9 **(Marks:1)** **Vu-Topper RM**

Dijkstra's algorithm :

A. Has greedy approach to find all shortest paths
B. Has both greedy and Dynamic approach to find all shortest paths
C. Has greedy approach to compute single source shortest paths to all other vertices **Page 154**
D. Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

Question No:10 **(Marks:1)** **Vu-Topper RM**

In Dijkstra's algorithm, initially the estimated value from source vertex to any vertex v is:

A. One (1)
B. Zero (0)
C. Infinity (∞) **Page 154**
D. Minus one (-1)

Question No:11 **(Marks:1)** **Vu-Topper RM**

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

From the given option's which one is correct regarding the time complexity of Dijkstra's algorithm:

A. O(N)
B. **O(N^2)** **Google**
C. O(N^3)
D. O(log N)

Question No:12 **(Marks:1)** **Vu-Topper RM**

In Bellman-Ford Algorithm, relaxation applies to _____ of the graph.

A. Every edge **Page 159**
B. Every Vertices
C. Only First edge
D. Only First Vertices

Question No:13 **(Marks:1)** **Vu-Topper RM**

The key[u] is the weight of the _____ going from u to any vertex in S.

A. Edge
B. lighter edge
C. lightest edge **Page 151**
D. heavier edge

Question No:14 **(Marks:1)** **Vu-Topper RM**

In Bellman-Ford Algorithm, path consists of at most _____ edges.

A. V-1 **Page 160**
B. E+1
C. V+1
D. E-1

Question No:15 **(Marks:1)** **Vu-Topper RM**

Overall time for Kruskal algorithm is:

A. Θ(logE)
B. Θ(VlogE)
C. Θ(ElogV) **Page 149**
D. Θ(ElogE)

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:16 **(Marks:1)** **Vu-Topper RM**

A graph may contain ----------.

A. No MST
B. One or zero MST
C. Exactly one MST
D. More than one MST **Google**

Question No:17 **(Marks:1)** **Vu-Topper RM**

In which algorithm, information of shortest path is propagated sequentially along each shortest path in the graph.

A. Prim's
B. Dijkstra's
C. Bellman-Ford **Page 16**
D. Brute-force technique

Question No:18 **(Marks:1)** **Vu-Topper RM**

Edge weights can be interpreted as distance _____.

A. in breadth-First Search
B. in Queue's
C. in the shortest-paths **Page 153**
D. in depth-First Search

Question No:19 **(Marks:1)** **Vu-Topper RM**

In the shortest-paths problem, we are given a weighted of_____G = (V, E).

A. Line graph
B. Directed graph **Page 153**
C. Weighted graph
D. Un-directed graph

Question No:20 **(Marks:1)** **Vu-Topper RM**

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the _____ has relatively few _____.

A. tree, edges
B. graph, edges **Google**

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

C. tree, branches
D. graph, branches

Question No:21 (Marks:1) Vu-Topper RM
From given algorithms which one considered as best for finding the shortest-path:
A. BFS
B. DFS
C. Dijkstra's algorithm Page 154
D. Bellman-Ford algorithm

Question No:22 (Marks:1) Vu-Topper RM
Finding the faster result of the shortest path from u to v for every pair of vertices u and v we use_____.
A. both I and II
B. All-pairs shortest-paths problem Page 153
C. Two-pairs shortest-paths problem
D. Single-pairs shortest-paths problem

Question No:23 (Marks:1) Vu-Topper RM
The running time of Bellman-Ford Algorithm is _____.
A. Θ(VE) Page 159
B. Θ(E + E)
C. Θ(V + E)
D. Θ(V + V)

Question No:24 (Marks:1) Vu-Topper RM
Dijkstra's algorithm is a simple_____ algorithm for computing the single-source shortest-paths to all other vertices.
A. Greedy Page 154
B. Brute-Force
C. Bellman-Ford
D. Divide and conquer

Question No:25 (Marks:1) Vu-Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

In Prim's algorithm, we will make use of_____.

A. List
B. Array
C. Stack
D. Priority Queue Page 150

Question No:26 (Marks:1) Vu-Topper RM

What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

A. (log E)
B. (V)
C. (V+E)
D. O(log V) Page 152

Question No:27 (Marks:1) Vu-Topper RM

Bellman-Ford Algorithm does not allow G(graph) to have _____.

A. positive cost cycles
B. negative cost cycles Page 159
C. positive weights edges
D. negative weights edges

Question No:28 (Marks:1) Vu-Topper RM

Which of the following statement is false about Dijkstra's Algorithm?

A. It works on a weighted directed graph
B. It is used to solve Single-source shortest path
C. It can be applied on graphs having a negative weight function
D. Its implementation in data structure is possible through the priority queue

Question No:29 (Marks:1) Vu-Topper RM

In _____ algorithm(s), at any time, the subset of edges A forms a single tree.

A. BFS
B. Prim's Page 149

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

C. Kruskal's
D. kruskal's and Prim's

Question No:30 (Marks:1) Vu-Topper RM
Floyd-Warshall Algorithm is based on _____.
A. Complexity theory
B. Greedy Approach
C. Divide and Conquer
D. Dynamic Programming Page 161

Question No:31 (Marks:1) Vu-Topper RM
Dijkstra's Algorithm cannot be applied on _____.
A. unweighted graphs
B. directed and weighted graphs
C. undirected and unweighted graphs
D. graphs having negative weight function

Question No:32 (Marks:1) Vu-Topper RM
Bellman-Ford algorithm is slower than_____.
A. Prim's
B. Dijkstra's Page 159
C. Graph Algorithm
D. Brute-force technique

Question No:33 (Marks:1) Vu-Topper RM
In Bellman-Ford Algorithm, relaxation applies to every edge of the graph and repeat this _____ time.
A. V-1 Page 159
B. E-1
C. V+1
D. E+1

Question No:34 (Marks:1) Vu-Topper RM
Which of the following is used in the data structure for implementing Dijkstra's Algorithm?
A. Stack's
B. Max heap

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

C. Priority queue **Google**
D. Circular queue

Question No:35 **(Marks:1)** **Vu-Topper RM**
In Prim's algorithm, If there is no edge from u to a vertex in S, we set the key value to_____.
A. 0
B. -1
C. ∞ **Page 151**
D. 1

Question No:36 **(Marks:1)** **Vu-Topper RM**
Prim's algorithm is based on ---------- strategy.
A. Greedy **Page 150**
B. Exponential
C. Divide and Conquer
D. Dynamic programming

Question No:37 **(Marks:1)** **Vu-Topper RM**
Dijkstra's Algorithm is used to solve_____problems.
A. Sorting & searching
B. All-pair shortest path
C. Multi-source shortest path
D. Single-source shortest path **Page 154**

Question No:38 **(Marks:1)** **Vu-Topper RM**
The process of updating estimates in Dijkstra's algorithm is called_____.
A. Insertion
B. Updating
C. Relaxation **Page 154**
D. Amendment

Question No:39 **(Marks:1)** **Vu-Topper RM**

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

_____is commonly the running time of Dijkstra's Algorithm using the binary heap method.

A. Θ(log E)
B. Θ(Vlog)
C. Θ(E log V) **Page 156**
D. Θ(Blog V)

Question No:40 **(Marks:1)** **Vu-Topper RM**

Kruskal's algorithm works by adding _____ in increasing order of weight (lightest edge first).

A. Trees
B. Edges **Page 147**
C. Vertices
D. Weights

Question No:41 **(Marks:1)** **Vu-Topper RM**

Dijkstra's algorithm works on a weighted directed graph G =(V, E) in which all_____weights are non-negative.

A. links
B. Edges **Page 154**
C. Nodes
D. Vertices

Question No:42 **(Marks:1)** **Vu-Topper RM**

In Kruskal's algorithm, the next _____ is not added to viable set A, if its adding induce a/an cycle.

A. Tree
B. Edge
C. Cycle **Page 147**
D. Vertex

Question No:43 **(Marks:1)** **Vu-Topper RM**

The breadth-first-search algorithm is a shortest-path algorithm that works on_____graphs.

A. Directed
B. Weighted
C. Un-directed

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

D. Un-weighted **Page 153**

Question No:44 **(Marks:1)** **Vu-Topper RM**

An un-weighted graph can be considered as a graph in which every edge has weight_____unit.

A. 1 **Page 153**

B. 3

C. 5

D. 7

Question No:45 **(Marks:1)** **Vu-Topper RM**

The tricky part of _____ algorithm(s) is/are, how to detect whether the addition of an edge will create a cycle in viable set A.

A. DFS

B. Prim's

C. Kruskal's **Page 159**

D. Both Krusal's and Prim's

Question No:46 **(Marks:1)** **Vu-Topper RM**

As the Kruskal's algorithm runs, the edges in viable set A induce a _____ on the vertices.

A. Set

B. Tree

C. Graph

D. Forest **Page 147**

Question No:47 **(Marks:1)** **Vu-Topper RM**

In Dijkstra’s algorithm the estimated value of source vertex d[s] is:

A. Equal to 1

B. Equal to 0 **Page 154**

C. Greater than 0

D. Greater than 1

Question No:48 **(Marks:1)** **Vu-Topper RM**

Equivalence relation partitions the vertices into _____ classes of mutually reachable vertices and these are the strong components

A. Variance

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What’s app 03224021365

B. Non classes
C. Equivalence **Page 136**
D. Non equivalence

Question No:49 **(Marks:1)** **Vu-Topper RM**
The ancestor and descendent relation can be nicely inferred by the _____ lemma.
A. Node
B. Division
C. Addition
D. Parenthesis **Page 129**

Question No:50 **(Marks:1)** **Vu-Topper RM**
Networks are _____ in the sense that it is possible from any location in the network to reach any other location in the digraph.
A. Not graphs.
B. Complete **Page 135**
C. Incomplete
D. Transportation

Question No:51 **(Marks:1)** **Vu-Topper RM**
In strong components algorithm, first of all DFS is run for getting _____ times of vertices.
A. Start **Page 138**
B. Finish
C. Middle
D. Both start & finish

Question No:52 **(Marks:1)** **Vu-Topper RM**
Timestamp structure of _____ is used in determining the strong components of a digraph.
A. DFS **Google**
B. BFS
C. MST
D. Both DFS, BFS

Question No:53 **(Marks:1)** **Vu-Topper RM**

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

In Timestamped DFS-cycles lemma, if edge (u, v) is a tree, forward or cross edge, then _____

A. f[u]>f[v]
B. f[u]<f[v]
C. f[u]⩽f[v]
D. f[u]⩾f[v]

Question No:54 **(Marks:1)** **Vu-Topper RM**
The component digraph is necessarily _____.

A. Cyclic
B. Strong
C. Acyclic **Page 136**
D. Straight

Question No:55 **(Marks:1)** **Vu-Topper RM**
In computing the _____ components of a digraph, vertices of the digraph are partitioned into subsets.

A. Best
B. Worst
C. Weakly connected
D. Strongly connected **Page 135**

Question No:56 **(Marks:1)** **Vu-Topper RM**
There exist a unique path between any _____ vertices of a free tree.

A. One
B. Two
C. five
D. three

Question No:57 **(Marks:1)** **Vu-Topper RM**
The _____ given by DFS allow us to determine whether the graph contains any cycles.

A. Order
B. Time stamps **Page 130**
C. BFS traversing
D. Topological sort

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:58 **(Marks:1)** **Vu-Topper RM**

Forward edge is:

A. (u, v) where u is a proper descendent of v in the tree.

B. (u, v) where v is a proper descendent of u in the tree.
Page 129

C. (u, v) where v is a proper ancesstor of u in the tree.

D. (u, v) where u is a proper ancesstor of v in the tree.

Question No:59 **(Marks:1)** **Vu-Topper RM**

Once you enter a strong component, every vertex in thecomponent is _____.

A. Removed

B. Reachable **Page 137**

C. not reachable

D. reachable some times

Question No:60 **(Marks:1)** **Vu-Topper RM**

In Generic approach determining of Greedy MST, we maintain a subset A of _____ .

A. Paths

B. Edges **Page 143**

C. Cycles

D. Vertices

Question No:61 **(Marks:1)** **Vu-Topper RM**

If you find yourself in maze the better traversel approach will be :

A. DFS

B. BFS

C. Level order

D. DFS

Question No:62 **(Marks:1)** **Vu-Topper RM**

Computing the strongly connected components of a digraph is a/an _____ of the problem to determine whether a digraph is strongly connected or not.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

A. Size
B. Connection
C. Optimization
D. Generalization **Page 135**

Question No:63 **(Marks:1)** **Vu-Topper RM**
A free tree with n _____ have exactly n-1 _____.
A. vertices,edges **Page 142**
B. vertices,nodes
C. nodes,vertices
D. vertices,nodes

Question No:64 **(Marks:1)** **Vu-Topper RM**
_____ components are not affected by reversal of all edges in terms of vertices reachability.
A. First two
B. Last two
C. Weakly connected
D. Strongly connected **Page 139**

Question No:65 **(Marks:1)** **Vu-Topper RM**
Adding any edge to a free tree creates a unique _____ .
A. Vertex
B. Cycle **Page 142**
C. Edge
D. Strong component

Question No:66 **(Marks:1)** **Vu-Topper RM**
In computing the strongly connected components of a digraph, vertices of the digraph are _____ into subsets.
A. Joined
B. Deleted
C. Created
D. Partitioned **Page 135**

Question No:67 **(Marks:1)** **Vu-Topper RM**
For _____ graphs, there is no distinction between forward and back edges.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

A. Large
B. Medium
C. Directed
D. Undirected **Page 130**

Question No:68 **(Marks:1)** **Vu-Topper RM**
A topological sort of a DAG is a _____ ordering of the vertices of the DAG such that for each edge (u, v), u appears before v in the ordering.
A. Linear **Page 134**
B. Parallel
C. Sequence
D. Non-Linea

Question No:69 **(Marks:1)** **Vu-Topper RM**
By breaking any edge on a cycle created in free tree, the free _____ is restored.
A. Tree **Page 142**
B. Edge
C. Cycle
D. Vertex

Question No:70 **(Marks:1)** **Vu-Topper RM**
Which technique is used in the implementation of Kruskal solution for the MST?
A. Greedy Technique **Page 142**
B. Divide-and-Conquer Technique
C. Dynamic Programming Technique
D. The algorithm combines more than one of the above techniques i.e. Divide-and-Conquer and Dynamic Programming

Question No:71 **(Marks:1)** **Vu-Topper RM**
A strongly connected component only apply to:
A. Directed Graph **Page 135**
B. Undirected Graph
C. Breadth First Search
D. Minimum Spanning Tree

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:72 **(Marks:1)** **Vu-Topper RM**

The relationship between number of back edges and number of cycles in DFS is,

A. Both are equal
B. Back edges are half of cycles
C. Back edges are one quarter of cycles
D. There is no relationship between no. of edges and cycles
Page 131

Question No:73 **(Marks:1)** **Vu-Topper RM**

Digraphs _____ in communication and transportation networks.

A. are used **Page 135**
B. are not used
C. parts are used
D. final value is used

Question No:74 **(Marks:1)** **Vu-Topper RM**

Back edge is:

A. (u, v) where v is an ancestor of u in the tree. **Page 128**
B. (u,v) where u is an ancesstor of v in the tree.
C. (u, v) where v is an predcessor of u in the tree.
D. None of above

Question No:75 **(Marks:1)** **Vu-Topper RM**

There are no _____ edges in undirected graph.

A. Back
B. Both
C. Cross **Page 130**
D. Forward

Question No:76 **(Marks:1)** **Vu-Topper RM**

In strong components algorithm, the form of graph is used in which all the _____ of original graph G have been reversed in direction.

A. Edges **Page 138**
B. Trees
C. Vertices

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

D. Both edges & vertices

Question No:77 (Marks:1) Vu-Topper RM

Which activity creates a unique cycle in a free tree:

A. adding root

B. adding any edge Page 142

C. adding any vertex

D. adding any sub tree

Question No:78 (Marks:1) Vu-Topper RM

You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T.?

A. (V + E) Page 138

B. (V E)

C. (V)

D. (V^2)

Question No:79 (Marks:1) Vu-Topper RM

The time complexity to compute Graph transpose G^T is (V+E), if you have _____ for G.

A. Stack

B. Array list

C. Complete list

D. An adjacency list Page 138

Question No:80 (Marks:1) Vu-Topper RM

In undirected graph, by convention all the edges are called _____ edges.

A. Back Page 130

B. Cross

C. Forward

D. Both forward and back

Question No:81 (Marks:1) Vu-Topper RM

The _____ given by DFS allow us to determine a number of things about a graph or digraph.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

A. line stamps
B. node stamps
C. color stamps
D. time stamps **Page 130**

Question No:82 **(Marks:1)** **Vu-Topper RM**
If a subset of edges A is viable for building MST, it can not contain a/an _____.
A. Edge
B. Cycle **Page 143**
C. Graph
D. Vertex

Question No:83 **(Marks:1)** **Vu-Topper RM**
A free tree with n vertices have exactly _____ edges.
A. 1
B. N
C. n-1 **Page 142**
D. n+1

Question No:84 **(Marks:1)** **Vu-Topper RM**
_____ technique is look like propagating wave-front outward.
A. Generic Traversal
B. Depth First Traversal
C. Time Stamp Traversal
D. Breath first traversal **Page 117**

Question No:85 **(Marks:1)** **Vu-Topper RM**
In Timestamped DFS, If there is a back edge (u, v) then v is an ancestor of u and by following tree edge from v to u, we get _____.
A. a line
B. a cycle **Page 131**
C. a graph
D. nothing

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:86 **(Marks:1)** **Vu-Topper RM**

In strong components algorithm, vertices are sorted in_____ order of finish times.

A. Any
B. Strong
C. Increasing
D. Decreasing **Page 141**

Question No:87 **(Marks:1)** **Vu-Topper RM**

A fully connected undirected graph of 5 nodes will have _____ edges.

A. 4
B. 5
C. 10
D. 15

Question No:88 **(Marks:1)** **Vu-Topper RM**

A digraph is strongly connected under what condition?

A. A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v .
B. A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa.
C. A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.
D. A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

Question No:89 **(Marks:1)** **Vu-Topper RM**

We say that two vertices u and v are mutually _____ if u can reach v and vice versa.

A. Crossed
B. Forward
C. Reachable **Page 135**
D. Not Reachable

Question No:90 **(Marks:1)** **Vu-Topper RM**

In Timestamped DFS, No back edges means _____.

A. DFS

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

B. BFS
C. 1 cycle
D. no cycles **Page 131**

Question No:91 **(Marks:1)** **Vu-Topper RM**
Cross edge is :
A. (u, v) where u and v are not ancestor of one another
B. (u, v) where u is ancesstor of v and v is not descendent of u.
C. (u, v) where u and v are not ancestor or descendent of one another
D. (u, v) where u and v are either ancestor or descendent of one another.

Question No:92 **(Marks:1)** **Vu-Topper RM**
In digraph G=(V,E) ;G has cycle if and only if
A. The DFS forest has forward edge.
B. The DFS forest has back edge **Page 131**
C. The DFS forest has both back and forward edge
D. BFS forest has forward edge

Question No:1 **(Marks:1)** **Vu-Topper RM**
If we encode and compress text using ASCII standard each character is represented b
A. Fixed length code word of 4 bits
B. Variable length code word up to 4 bits
C. Fixed length code word of 8 bits
D. Variable length code word up to 8 bits

Question No:93 **(Marks:1)** **Vu-Topper RM**
G power T for graph can be computed in
A. Theta(VE)
B. Theta(V log E)
C. Theta(E log V)
D .Theta (V+E)

Question No:94 **(Marks:1)** **Vu-Topper RM**
Adding any edge to free tree

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

A. Keep it free tree and increase the size of the tree.
B. Create a unique cycle.
C. Not allow to add edge to free tree.
D. Create multiple cycles.

Question No:95 **(Marks:1)** **Vu-Topper RM**
Which sorting algorithn is faster :
A. O(n^2)
B. O(nlogn)
C. O(n+k)
D. O(n^3)

Question No:96 **(Marks:1)** **Vu-Topper RM**
Networks are complete in the sense that it is possible from any location in the network to reach any other location in the digraph.
True
False

Question No:97 **(Marks:1)** **Vu-Topper RM**
Runtime complexity of Prim's algorithm is _____.
A. V log V
B. E log V
C. log V
D. None of the above

Question No:98 **(Marks:1)** **Vu-Topper RM**
Adding any edge to a free tree creates a unique cycle.
True
False

Question No:99 **(Marks:1)** **Vu-Topper RM**
For undirected graph, there is no distinction between forward and back edges.
True
False

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:100 **(Marks:1)** **Vu-Topper RM**

in strong components algorithm, first of all DFS is run for computing finish times of vertices.

True

False

Question No:101 **(Marks:1)** **Vu-Topper RM**

Which method is preferable for dealing with chain matrix multiplication?

A. Divide and conquer strategy

B. Dynamic programming formulation

C. Graph theory

D. Greedy Approach

Question No:102 **(Marks:1)** **Vu-Topper RM**

Huffman algorithm produces the…….prefix code tree.

A. Better

B. Optimal

C. Worst

D. Best

Question No:103 **(Marks:1)** **Vu-Topper RM**

A….w is adjacent to vertex v if there is an edge from v to w.

A. Acyclic

B. Vertex

C. Loop

D. Cycle

Question No:104 **(Marks:1)** **Vu-Topper RM**

Using ASCII standard the string “greedy” will be encoded with

A. 44 bits

B. 120 bits

C. 40 bits

D. 48 bits

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What’s app 03224021365

Question No:105 **(Marks:1)** **Vu-Topper RM**

In activity scheduling algorithm, each activity is represented by a

A. Rectangle

B. Square

C. Circle

D. Triangle

Question No:106 **(Marks:1)** **Vu-Topper RM**

Those problems in which greedy finds good, but not always best is called a greedy…..

A. **Heuristic**

B. Solution

C. Result

D. Algorithm

Question No:107 **(Marks:1)** **Vu-Topper RM**

The knapsack problem belongs to the domain of…..Problems

A. Searching

B. Sorting

C. Linear solution

D. Optimization

Question No:108 **(Marks:1)** **Vu-Topper RM**

The general coin change problem can be solved using

A. Recursion

B. Greedy algorithm

C. Dynamic programming

D. Divide and conquer

Question No:109 **(Marks:1)** **Vu-Topper RM**

Huffman algorithm generates an optimum.code

A. Postfix

B. Infix

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

C. None of the given options

D. Prefix

Question No:110 (Marks:1) Vu-Topper RM

..... ways of representing graphics

A. 1

B. 2

C. 3

D. 4

Question No:111 (Marks:1) Vu-Topper RM

Knapsack word originates from……language

A. German

B. English

C. French

D. Norwegian

Question No:112 (Marks:1) Vu-Topper RM

Graphs are important.....model for many application problems

A. Mathematical

B. Unpredictable

C. Haphazard

D. Unsystematic

Question No:113 (Marks:1) Vu-Topper RM

Which type of algorithm is harder to prove the correctness?

A. Dynamic

B. Greedy

C. Divide and conquer

D. Brute force

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:114 **(Marks:1)** **Vu-Topper RM**

..... Items are not allowed in 0/1 knapsack problem

A. Fractional

B. 0

C. 1

D. 0/1

Question No:115 **(Marks:1)** **Vu-Topper RM**

Matrix multiplication is a(n) .....operation

A. Nether commutative nor associative

B. Transitive

C. Commutative

D. Associative

Question No:116 **(Marks:1)** **Vu-Topper RM**

For a Diagraph G= (V.E), Sum of in-degree (v) --.

A. Not equal to sum of out-degree(v)

B. = sum of out-degree(v)

C. < sum of out-degree(v)

D. > sum of out-degree(v)

Question No:117 **(Marks:1)** **Vu-Topper RM**

DFS or BFS yields a of the graph.

A. Traversed tree

B. Spanning tree **Page 125**

C. Simple tree

D. Free tree

Question No:118 **(Marks:1)** **Vu-Topper RM**

Using ASCII code, each character is represented by a fixed-length code of bits per character.

A. 4

B. 6

C. 10

D. 8

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:119 **(Marks:1)** **Vu-Topper RM**

In Knapsack Problem, the goal is to put items in the Knapsack such that the value of the items is----------subject to weight limit of the Knapsack.

A. Minimized
B. Decreased
C. Maximized
D. None of the above given

Question No:120 **(Marks:1)** **Vu-Topper RM**

A graph is said to be acyclic if it contains --.

A. At least one cycle
B. Exactly one cycle
C. Always more than one cycle
D. No cycles

Question No:121 **(Marks:1)** **Vu-Topper RM**

The number of edges that come out of a vertex is called the of that vertex in the digraph.

A. Post-degree
B. in-degree
C. out-degree
D. pre-degree

Question No:122 **(Marks:1)** **Vu-Topper RM**

What general property of the list indicates that the graph has an isolated vertex?

A. There is Null pointer at the end of list.
B. The Isolated vertex is not handled in list.
C. Only one value is entered in the list.
D. There is at least one null list.

Question No:123 **(Marks:1)** **Vu-Topper RM**

As the Kruskal's runs, the edges in viable set A induce a _____ on the vertices.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

A. Set
B. Graph
C. Tree
D. Fore

Question No:124 (Marks:1) Vu-Topper RM

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

A. 3x2
B. 2x3
C. 2x2
D. 3x3

Question No:125 (Marks:1) Vu-Topper RM

A/an is one in which you want to find, not just a solution, but the best solution.

A. Optimization problem
B. Divide and Conquer
C. NP complete problem
D. Best problem

Question No:126 (Marks:1) Vu-Topper RM

Fractional Knapsack is founded on--- method.

A. Greedy page
B. Recursive
C. Divide and Conquer
D. Dynamic programming

Question No:127 (Marks:1) Vu-Topper RM

Find the maximum value of the items which can carry using knapsack weight capacity

=50

ITEM WEIGHT	10	20	30	70

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

VALUE	70	20	80	200

A. 90
B. 280
C. 200
D. 100

Question No:128 (Marks:1) Vu-Topper RM

If the graph is represented using an adjacency matrix, then Breadth-first search takes----------time.

A. O(E+1)
B. O(V^2)
C. O(V)
D. O(E)

Question No:129 (Marks:1) Vu-Topper RM

In inductive approach of Knapsack problem, we consider 2 cases, ----------or --.Median, Mode

A. Recursive, Iterative
B. Leave object, Take object
C. Sequentially, Parallel

Question No:130 (Marks:1) Vu-Topper RM

A Greedy algorithm can NOT be used to solve all the problems.

A. Dynamic programming
B. Memorization programming
C. Edit-distance programming
D. Storing value programming

Question No:131 (Marks:1) Vu-Topper RM

In Huffman encoding, the is the number of occurrences of a character divided by the total characters in the message.

A. Counting
B. Parsing
C. Relative Probability
D. Weight

Question No:132 (Marks:1) Vu-Topper RM

The Binary Tree constructed by a Huffman Encoding is a:

A. Full Binary Tree

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

B. Partial Binary Tree
C. Incomplete Binary Tree
D. None of the given option

Question No:133 (Marks:1) Vu-Topper RM

Following is not the application of Edit Distance Problem.

A. Speech recognition
B. Spelling correction
C. Ascending order
D. Computational Molecular Biology

Question No:134 (Marks:1) Vu-Topper RM

Consider three Matrices X, Y, Z of dimensions 1x2, 2x3, 3x4 respectively. The numberof multiplication of(XYZ) is:

A. 18
B. 32
C. 24
D. 30

Question No:135 (Marks:1) Vu-Topper RM

In-----Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

A. 0
B. 1
C. 0/1
D. Fractional

Question No:136 (Marks:1) Vu-Topper RM

An in-place sorting algorithm is one that uses additional array for storage.

A. Always
B. Permanently
C. Does not
D. Sometime

Question No:137 (Marks:1) Vu-Topper RM

If Matrix-A has dimensions “pxq” and Matrix-B has dimensions

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What’s app 03224021365

"qxr", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

A. P x q
B. P x r
C. q x r
D. q x p

Question No:138 (Marks:1) Vu-Topper RM

Counting sort is suitable to sort the elements in range 1 to K.

A. K is large
B. K is small
C. K may be large or small
D. None

Question No:139 (Marks:1) Vu-Topper RM

When matrix A of 5x 3 is multiply with matrix B of 3x 4 then the multiplication required is:

A. 15
B. 12
C. 36
D. 60

Question No:140 (Marks:1) Vu-Topper RM

---------- is a linear time sorting algorithm.

A. Merge sort
B. Quick sort
C. Bubble sort
D. Radix sort

Question No:141 (Marks:1) Vu-Topper RM

In Dynamic Programming approach, we do not store the solution to each sub problem in case if it reappears.

True
False

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:142 **(Marks:1)** **Vu-Topper RM**

Dynamic Programming approach is usually useful in solving optimization problem.

True

False

Question No:143 **(Marks:1)** **Vu-Topper RM**

Which of the following algorithm provides an optimal solution for the activity selection problem?

A. Divide and Conquer
B. Brute force
C. Greedy
D. Recursive

Question No:144 **(Marks:1)** **Vu-Topper RM**

A graph is if every vertex can reach every other vertex.

A. Connected
B. Cycle
C. Acyclic
D. Loop

Question No:145 **(Marks:1)** **Vu-Topper RM**

In a Huffman encoding when a new node is created by combining two nodes, the new node is placed in the ___.

A. Priority queue
B. Linked list
C. Min heap tree
D. Graph traversal

Question No:146 **(Marks:1)** **Vu-Topper RM**

In___algorithm, you hope that by choosing a local optimum at each step, you will end up a global optimum.

A. Simple
B. Divide and conquer
C. Greedy
D. Brute Force

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:147 **(Marks:1)** **Vu-Topper RM**

The string “Imncde” is coded with ASCII code, the message length would bebits.

A. 24
B. 36
C. 48
D. 60

Question No:148 **(Marks:1)** **Vu-Topper RM**

For graph traversal, breadth-first search strategy __

A. Is always recursive
B. Cannot br recursive
C. Cannot be non-recursive
D. Can be both recursive and non-recursive

Question No:149 **(Marks:1)** **Vu-Topper RM**

In activity scheduling algorithm , the width of a rectangle __

A. Is always ignored
B. Directs towards recursion
C. Should be maximized
D. **Indicates the duration of an activity**

Question No:150 **(Marks:1)** **Vu-Topper RM**

If the graph is represented using an adjacency list, then Breadth-first search take time

A. O(V^2)
B. O(V)
C. O(V+E)
D.O(E+1)

Question No:151 **(Marks:1)** **Vu-Topper RM**

Suppose you are given infinite coins of 1,2 ,3, and 4.Select the ways of the minimum numberof coins that required to achieve a sum of 6:

A. 1
B. 2

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What’s app 03224021365

C. 3
D. 4

Question No:152 (Marks:1) Vu-Topper RM

sing ASCII standard the string “greedy” will be encoded with

A. 48 bitS
B. 20 bits
C. 44 bits
D.40 bits

Question No:153 (Marks:1) Vu-Topper RM

The Huffman codes provide a method of data efficiency.

A. Reading/Writing
B. Encoding/Decoding
B. Divide/Conquer
C. Inserting/Deleting

Question No:154 (Marks:1) Vu-Topper RM

In the context of activity selection algorithm, time s dominated by sorting of the activities by------.

A. Start Times
B. Finish Times
C. Average Times
D.CPU Burst Times

Question No:155 (Marks:1) Vu-Topper RM

Time complexity of the “0-1” knapsack algorithm depends on----

A. Number of items
B. Capacity of the knapsack
C. Size of the Table
D. Number of items and capacity of knapsack

Question No:156 (Marks:1) Vu-Topper RM

The greedy approach gives us an optimal solution when the coins are all powers of a --------denomination

A. Fixed
B. Variable

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What’s app 03224021365

C. Constant
D. Static

Question No:157 (Marks:1) Vu-Topper RM
In Activity Selection, we say that two activities are non-interfering if their start-finish interval overlap
A. Do
B. Do not.
C. Sometimes
D. Once

Question No:158 (Marks:1) Vu-Topper RM
How many steps are involved to design the dynamic programming strategy?
A. 2
B. 3
C. 1
D. 4

Question No:159 (Marks:1) Vu-Topper RM
Bag is a.....
A. type of algorithm
B. data structure
C. program
D. compiler

Question No:160 (Marks:1) Vu-Topper RM
If a problem is in NP-complete, it must also be in NP.
True
False

Question No:161 (Marks:1) Vu-Topper RM
The Huffman algorithm finds a optimal solution.
True
False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:162 **(Marks:1)** **Vu-Topper RM**
The Huffman algorithm finds an exponential solution
True
False

Question No:163 **(Marks:1)** **Vu-Topper RM**
The Huffman algorithm finds a polynomial solution
True
False

Question No:164 **(Marks:1)** **Vu-Topper RM**
The greedy part of the Huffman encoding algorithm is to first find two nodes withs sallest frequency.
True
False

Question No:165 **(Marks:1)** **Vu-Topper RM**
The code word assigned to characters by the Huffman algorithm have the property that no code word is the prefix of any other.
True
False

Question No:166 **(Marks:1)** **Vu-Topper RM**
Dijkestra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.
True
False

Question No:167 **(Marks:1)** **Vu-Topper RM**
The term "coloring" came form the original application which was in architectural design.
True
False

Question No:168 **(Marks:1)** **Vu-Topper RM**
In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

True
False

Question No:169 **(Marks:1)** **Vu-Topper RM**
Dijkstra's algorithm is operates by maintaining a subset of vertices
True
False

Question No:170 **(Marks:1)** **Vu-Topper RM**
We do sorting to,
A. keep elements in random positions
B. keep the algorithm run in linear order
C. keep the algorithm run in (log n) order
D. keep elements in increasing or decreasing order

Question No:171 **(Marks:1)** **Vu-Topper RM**
After partitioning array in Quick sort, pivot is placed in a position such that
A. Values smaller than pivot are on left and larger than pivot are on right
B. Values larger than pivot are on left and smaller than pivot are on right
C. Pivot is the first element of array
D. Pivot is the last element of array

Question No:172 **(Marks:1)** **Vu-Topper RM**
Merge sort is stable sort, but not an in-place algorithm
True
False

Question No:173 **(Marks:1)** **Vu-Topper RM**
A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p . r) total entries in C and each takes__ to compute.
A. O (q)

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365
Vu Topper RM

B. O (1)
C. O (n^2)
D. O (n^3)

Question No:174 (Marks:1) Vu-Topper RM

One of the clever aspects of heaps is that they can be stored in arrays without using any ---------.

A. Pointers
B. constants
C. variables
D. functions

Question No:175 (Marks:1) Vu-Topper RM

Merge sort requires extra array storage,

True
False

Question No:176 (Marks:1) Vu-Topper RM

The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.

True
False

Question No:177 (Marks:1) Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with _____bits.

A. 80
B. 160
C. 320
D. 100

Question No:178 (Marks:1) Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 160 bits.

True
False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:179 **(Marks:1)** **Vu-Topper RM**
Using ASCII standard the string abacdaacac will be encoded with 10 bytes.
True
False

Question No:180 **(Marks:1)** **Vu-Topper RM**
The greedy part of the Huffman encoding algorithm is to first find two nodes with character frequency
True
False

Question No:181 **(Marks:1)** **Vu-Topper RM**
Huffman algorithm uses a greedy approach to generate an prefix code T that minimizesth expected length B (T) of the encoded string.
True
False

Question No:182 **(Marks:1)** **Vu-Topper RM**
An optimization problem is one in which you want to find,
A. Not a solution
B. An algorithm
C. Good solution
D. The best solution

Question No:183 **(Marks:1)** **Vu-Topper RM**
Although it requires more complicated data structures, Prim's algorithm for a minimumspanning tree is better than Kruskal's when the graph has a large number of vert
True
False

Question No:184 **(Marks:1)** **Vu-Topper RM**
If a problem is in NP, it must also be in P.
A. True
B. False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

C. unknown

Question No:185 (Marks:1) Vu-Topper RM

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

A. Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)

B. Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2)

C. Lists require more space than matrices and they take longer to find the weight of an edge (v1,v2)

D. Lists require more space than matrices but are faster to find the weight of an edge (v1,V2)

Question No:186 (Marks:1) Vu-Topper RM

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

A. v edges.

B. v – e + 5 edges

C. v + e edges.

D. None of these

Question No:187 (Marks:1) Vu-Topper RM

Maximum number of vertices in a Directed Graph may be $|V^2|$

True

False

Question No:188 (Marks:1) Vu-Topper RM

The Huffman algorithm finds a (n) __solution.

A. Optimal

B. Non-optimal

C. Exponential

D. Polynomial

Question No:189 (Marks:1) Vu-Topper RM

Edge (u, v) is a forward edge if

A. u is a proper descendant of v in the tree

B. v is a proper descendant of u in the tree pg#129

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

C. None of these

Question No:190 (Marks:1) Vu-Topper RM

In counting sort, once we know the ranks, we simply_numbers to their final positions in an output array.

A. Delete
B. copy
C. Mark
D. arrange

Question No:191 (Marks:1) Vu-Topper RM

Dynamic programming algorithms need to store the results of intermediate sub-problems.

True
False

Question No:192 (Marks:1) Vu-Topper RM

_____is a graphical representation of an algorithm

A. notation
B. notation
C. Flowchart
D. Asymptotic notation

Question No:193 (Marks:1) Vu-Topper RM

Which of the following is calculated with **big o notation**?

A. Lower bounds
B. Upper bounds
C. Both upper and lower bound
D. Medium bounds

Question No:194 (Marks:1) Vu-Topper RM

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

A. The array elements form a heap
B. Elements in each half of the array are sorted amongst themselves.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

C. Elements in the first half of the array are less than or equal to elements in the second half of the array
D. None of the above

Question No:195 (Marks:1) Vu-Topper RM

What is the solution to the recurrence T(n) = T(n/2)+n, T(1) = 1

A. O(logn)
B. O(n)
C. O(nlogn)
D. O(2n)

Question No:196 (Marks:1) Vu-Topper RM

Consider the following Huffman Tree The binary code for the string TEA is

A. 00 010
B. 011 00 010
C. 10 00 110
D. 11 10 110

Question No:197 (Marks:1) Vu-Topper RM

A greedy algorithm does not work in phases.

True
False

Question No:198 (Marks:1) Vu-Topper RM

Can an adjacency matrix for a directed graph ever not be square in shape?

Yes
No

Question No:199 (Marks:1) Vu-Topper RM

One of the clever aspects of heaps is that they can be stored in arrays without using any_____.

A. Pointers
B. constants

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

C. variables
D. functions

Question No:200 (Marks:1) Vu-Topper RM
Non-optimal or greedy algorithm for money change takes___
A. O(k)
B. O(kN)
C. O(2k)
D. O(N)

Question No:201 (Marks:1) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with 320 bits.
True
False

Question No:202 (Marks:1) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with 100 bits.
True
False

Question No:203 (Marks:1) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with 32 bytes
True
False

Question No:204 (Marks:1) Vu-Topper RM
Huffman algorithm uses a greedy approach to generate an **antefix** code T thatminimizes the expected length B (T) of the encoded string.
True
False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:205 **(Marks:1)** **Vu-Topper RM**

Depth first search is shortest path algorithm that works on un-weighted graphs.

True

False

Question No:206 **(Marks:1)** **Vu-Topper RM**

Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

True

False

Question No:207 **(Marks:1)** **Vu-Topper RM**

Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

A. k

B. d^k

C. True

D. False

Question No:208 **(Marks:1)** **Vu-Topper RM**

The term coloring came from the original application which was in map drawing.

True

False

Question No:209 **(Marks:1)** **Vu-Topper RM**

In the clique cover problem, for two vertices to be in the same group, they must be_____each other.

A. Apart from

B. Far from

C. Near to

D. Adjacent to

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:210 **(Marks:1)** **Vu-Topper RM**

Fixed-length codes may not be efficient from the perspective of _____the total quantity of data.

A. Minimizing
B. Averaging
C. Maximizing
D. Summing

Question No:211 **(Marks:1)** **Vu-Topper RM**

In greedy algorithm, at each phase, you take the_you can get right now, without regard for future consequences.

A. Worst
B. Minimum
C. Good
D. Best

Question No:212 **(Marks:1)** **Vu-Topper RM**

The difference between Prim s algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a same key.

True
False

Question No:213 **(Marks:1)** **Vu-Topper RM**

If there are n items, there are___possible combinations of the items.

A. 2
B. N
C. 2^n
D. 3^n

Question No:214 **(Marks:1)** **Vu-Topper RM**

In Knapsack Problem, the thief's goal is to put items in the bag such that the ___of the items does not exceed the limit of the bag.

A. Value
B. Weight
C. Length
D. Balance

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:215 **(Marks:1)** **Vu-Topper RM**
The knapsack problem does not belong to the domain of optimization problems.
True
False

Question No:216 **(Marks:1)** **Vu-Topper RM**
In Huffman encoding, for a given message string, the frequency of occurrence(relative probability) of each character in the message is determined last.
True
False

Question No:217 **(Marks:1)** **Vu-Topper RM**
Fixed-length codes are known for easy break up of a string into its individual characters.
True
False

Question No:218 **(Marks:1)** **Vu-Topper RM**
In Knapsack Problem, value and weight both are to be under consideration.
True
False

Question No:219 **(Marks:1)** **Vu-Topper RM**
Multiplication is____.
A. log n
B. n
C. n2
D. n3

Question No:220 **(Marks:1)** **Vu-Topper RM**
In DP based solution of knapsack problem, to compute entries of V we will imply a/an_approach.
A. Subjective
B. Inductive

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

C. Brute force
D. Combination

Question No:221 (Marks:1) Vu-Topper RM
A greedy algorithm sometimes works well for optimization problems.
True
False

Question No:222 (Marks:1) Vu-Topper RM
In Huffman encoding, frequency of each character can be determined by parsing themessage and____how many times each character (or symbol) appears.
A. Printing
B. Incrementing
C. Counting
D. Deleting

Question No:223 (Marks:1) Vu-Topper RM
Greedy algorithm can do very poorly for some problems.
True
False

Question No:224 (Marks:1) Vu-Topper RM
The Huffman codes provide a method of__data efficiently.
A. Reading
B. Encoding
C. Decoding
D. Printing

Question No:225 (Marks:1) Vu-Topper RM
In___based solution of knapsack problem, we consider 2 cases, Leave objectOr Take object.
A. Brute force
B. Dynamic programming

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:226 **(Marks:1)** **Vu-Topper RM**
In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.
True
False

Question No:227 **(Marks:1)** **Vu-Topper RM**
_____problem, we want to find the best solution.
A. Minimization
B. Averaging
C. Optimization
D. Maximization

Question No:228 **(Marks:1)** **Vu-Topper RM**
Counting Money problem is an example which cannot be optimally solved by greedy algorithm.
True
False

Question No:229 **(Marks:1)** **Vu-Topper RM**
Huffman algorithm generates an optimum prefix code.
True
False

Question No:230 **(Marks:1)** **Vu-Topper RM**
If the string "lmncde" is coded with ASCII code, the message length would be ____bits.
A. 24
B. B36
C. 48 **6*8=48**
D. 60

Question No:231 **(Marks:1)** **Vu-Topper RM**
There are_nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.
A. 2
B. 3
C. 4

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:232 **(Marks:1)** **Vu-Topper RM**

A number of lectures are to be given in a single lecture hall. Optimum scheduling forthis is an example of Activity selection.

True

False

Question No:233 **(Marks:1)** **Vu-Topper RM**

The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an____solution.

A. Simple
B. Sub optimal
C. Optimal
D. Non optimal

Question No:234 **(Marks:1)** **Vu-Topper RM**

The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

True

False

Question No:235 **(Marks:1)** **Vu-Topper RM**

An application problem is one in which you want to find, not just a solution, but the ___solution.

A. Simple
B. Good
C. Best

Question No:236 **(Marks:1)** **Vu-Topper RM**

Suppose that a graph G = (V,E) is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?

A. O(|V |^2)
B. O(|V | |E|)
C. O(|V |^2|E|)
D. O(|v|+|E|)

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:237 **(Marks:1)** **Vu-Topper RM**

Which is true statement?

A. **Breadth first search is shortest path algorithm that works on un-weighted graphs**

B. Depth first search is shortest path algorithm that works on un-weighted graphs.

C. Both of above are true.

D. None of above are true.

Question No:238 **(Marks:1)** **Vu-Topper RM**

Using ASCII standard the string “abacdaacacwe” will be encoded with _____ bits

A. 64

B. 128

C. 96

D. 120

Question No:239 **(Marks:1)** **Vu-Topper RM**

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

True

False

Question No:240 **(Marks:1)** **Vu-Topper RM**

What algorithm technique is used in the implementation of Kruskal solution for the MST?

A. **Greedy Technique**

B. Divide-and-Conquer Technique

C. Dynamic Programming Technique

D. The algorithm combines more than one of the above techniques.

Question No:241 **(Marks:1)** **Vu-Topper RM**

There is relationship between number of back edges and number of cycles in DFS

A. Both are equal.

B. Cycles are half of back edges.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What’s app 03224021365

C. Cycles are one fourth of back edges.
D. There is no relationship between back edges and number of cycles.

Question No:242 (Marks:1) Vu-Topper RM
In in-place sorting algorithm is one that uses arrays for storage:
A. An additional array
B. No additional array
C. Both of above may be true according to algorithm.
D. More than 3 arrays of one dimension.

Question No:243 (Marks:1) Vu-Topper RM
In stable sorting algorithm
A. One array is used
B. In which duplicating elements are not handled.
C. More than one arrays are required.
D. Duplicating elements remain in same relative position after sorting.

Question No:244 (Marks:1) Vu-Topper RM
Which sorting algorithm is faster :
A. O(n^2)
B. O(nlogn)
C. O(n+k)
D. O(n^3)

Question No:245 (Marks:1) Vu-Topper RM
In Quick sort algorithm, constants hidden in T(n lg n) are
A. Large
B. Medium
C. Not known
D. Small

Question No:246 (Marks:1) Vu-Topper RM
Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:
A. There is explicit combine process as well to conquer the solution.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

B. No work is needed to combine the sub-arrays, the array is already sorted.
C. Merging the sub arrays
D. None of the above

Question No:247 (Marks:1) Vu-Topper RM
Which may be stable sort:
A. Bubble sort
B. Insertion sort
C. Selection sort
D. Both of above

Question No:248 (Marks:1) Vu-Topper RM
In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent __ series in the analysis,
A. linear
B. arithmetic
C. exponent
D. Geometric

Question No:249 (Marks:1) Vu-Topper RM
Which may be stable sort:
A. Bubble sort
B. Insertion sort
C. Selection sort
D. Both of above

Question No:250 (Marks:1) Vu-Topper RM
In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent __ series in the analysis,
A. linear
B. arithmetic
C. exponent
D. Geometric

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:251 **(Marks:1)** **Vu-Topper RM**

How much time merge sort takes for an array of numbers?

A. T(n^2)

B. T(n)

C. T(log n)

D. T(n log n)

Question No:252 **(Marks:1)** **Vu-Topper RM**

We do sorting to,

A. keep elements in random positions.

B. keep the algorithm run in linear order.

C. keep the algorithm run in (log n) order.

D. keep elements in increasing or decreasing order.

Question No:253 **(Marks:1)** **Vu-Topper RM**

Dynamic programming algorithms need to store the results of intermediate sub-problems.

True

False

Question No:254 **(Marks:1)** **Vu-Topper RM**

Dijkstras single source shortest path algorithm works if all edges weights are non negative and there are no negative cost cycles.

False

True

Question No:255 **(Marks:1)** **Vu-Topper RM**

Dijkestra s single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.

True

False

Question No:256 **(Marks:1)** **Vu-Topper RM**

In the clique cover problem, for two vertices to be in the same group, they must be__each other.

A. Apart from

B. Far from

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

C. Near to
D. **Adjacent to**

Question No:257 (Marks:1) Vu-Topper RM
Fixed-length codes may not be efficient from the perspective of _____the total quantity of data.

A. Minimizing
B. Averaging
C. Maximizing
D. Summing

Question No:258 (Marks:1) Vu-Topper RM
In___based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.
Brute force
Dynamic programming

Question No:259 (Marks:1) Vu-Topper RM
In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.
True
False

Question No:260 (Marks:1) Vu-Topper RM
_____problem, we want to find the best solution.
A. Minimization
B. Averaging
C. Optimization
D. Maximization

Question No:261 (Marks:1) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with 10 bytes.
True
False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:262 **(Marks:1)** **Vu-Topper RM**

How many elements do we eliminate in each time for the Analysis of Selection algorithm?

A. (n / 2) + n elements
B. n / 4 elements
C. 2 n elements
D. n / 2 elements

Question No:263 **(Marks:1)** **Vu-Topper RM**

Slow sorting algorithms run in,

A. T(n^2)
B. T(n)
C. T(log n)
D. T(n log n)

Question No:264 **(Marks:1)** **Vu-Topper RM**

Counting sort is suitable to sort the elements in range 1 to k:

A. K is large
B. K is small
C. K may be large or small
D. None

Question No:265 **(Marks:1)** **Vu-Topper RM**

Heaps can be stored in arrays without using any pointers; this is due to the_nature of the binary tree,

A. left-complete
B. right-complete
C. tree nodes
D. tree leaves

Question No:266 **(Marks:1)** **Vu-Topper RM**

A heap is a left-complete binary tree that conforms to the __

A. increasing order only
B. heap order
C. decreasing order only
D. (log n) order

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:267 **(Marks:1)** **Vu-Topper RM**

Divide-and-conquer as breaking the problem into a small number of

A. pivot
B. Sieve
C. smaller sub problems
D. Selection

Question No:268 **(Marks:1)** **Vu-Topper RM**

In Sieve Technique we do not know which item is of interest

True
False

Question No:269 **(Marks:1)** **Vu-Topper RM**

The recurrence relation of Tower of Hanoi is given below T(n)={1 if n=1 and2T(n-1) if n >1 In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

A. 16
B. 10
C. 32
D. 31

Question No:270 **(Marks:1)** **Vu-Topper RM**

For the heap sort, access to nodes involves simple ___ operations.

A. arithmetic
B. binary
C. algebraic
D. logarithmic

Question No:271 **(Marks:1)** **Vu-Topper RM**

For the sieve technique we solve the problem,

A. recursively
B. mathematically
C. precisely
D. accurately

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:272 **(Marks:1)** **Vu-Topper RM**

The sieve technique works in___as follows

A. phases
B. numbers
C. integers
D. routines

Question No:273 **(Marks:1)** **Vu-Topper RM**

A (an)____is a left-complete binary tree that conforms to the heap order

A. heap
B. binary tree
C. binary search tree
D. array

Question No:274 **(Marks:1)** **Vu-Topper RM**

The sieve technique is a special case, where the number of sub problems is just

A. 5
B. Many
C. 1
D. Few

Question No:275 **(Marks:1)** **Vu-Topper RM**

Analysis of Selection algorithm ends up with,

A. T(n)
B. T(1 / 1 + n)
C. T(n / 2)
D. T((n / 2) + n)

Question No:276 **(Marks:1)** **Vu-Topper RM**

For the heap sort we store the tree nodes in

A. level-order traversal
B. in-order traversal
C. pre-order traversal
D. post-order traversal

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:277 **(Marks:1)** **Vu-Topper RM**

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

A. divide-and-conquer
B. decrease and conquer
C. greedy nature
D. 2-dimension Maxima

Question No:278 **(Marks:1)** **Vu-Topper RM**

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of __

A. n items
B. phases
C. pointers
D. constant

Question No:279 **(Marks:1)** **Vu-Topper RM**

Memorization is?

A. To store previous results for future use
B. To avoid these unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later
C. To make the process accurate
D. None of the above

Question No:280 **(Marks:1)** **Vu-Topper RM**

Quick sort is

A. Not stable but in place
B. Stable & in place
C. Stable but not in place
D. Sometime stable & some times in place

Question No:281 **(Marks:1)** **Vu-Topper RM**

One example of in place but not stable algorithm is

A. Merger Sort
B. Quick Sort
B. Continuation Sort

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

C. Bubble Sort

Question No:282 **(Marks:1)** **Vu-Topper RM**

Continuation sort is suitable to sort the elements in range 1 to k

A. K is Large
B. K is not known
C. K may be small or large
D. K is small.

Question No:283 **(Marks:1)** **Vu-Topper RM**

Which may be a stable sort?

A. Merger
B. Insertion
C. None of the above
D. Both above

Question No:284 **(Marks:1)** **Vu-Topper RM**

An in place sorting algorithm is one that uses___arrays for storage

A. Two dimensional arrays
B. More than one array
C. No Additional Array
D. None of the above

Question No:285 **(Marks:1)** **Vu-Topper RM**

Continuing sort has time complexity of ?

A. O(n)
B. O(n+k)
C. O(nlogn)
D. O(k)

Question No:286 **(Marks:1)** **Vu-Topper RM**

single item from a larger set of _

A. phases
B. n items
C. pointers
D. constant

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:287 **(Marks:1)** **Vu-Topper RM**

For the Sieve Technique we take time

A. T(nk)
B. T(n / 3)
C. n^2
D. n/3

Question No:288 **(Marks:1)** **Vu-Topper RM**

Due to left complete nature of binary tree, the heap can be stored in

A. Arrays
B. Structures
C. Link List
D. Stack

Question No:289 **(Marks:1)** **Vu-Topper RM**

What type of instructions Random Access Machine (RAM) can execute?

A. Algebraic and logic
B. Geometric and arithmetic
C. Arithmetic and logic
D. Parallel and recursive

Question No:290 **(Marks:1)** **Vu-Topper RM**

What is the total time to heapify?

A. O(log n)
B. O(n log n)
C. $O(n^2 \log n)$
D. $O(\log^2 n)$

Question No:291 **(Marks:1)** **Vu-Topper RM**

word Algorithm comes from the name of the muslim author _____

Abu Ja'far Mohammad ibn Musa al-Khowarizmi.

Question No:292 **(Marks:1)** **Vu-Topper RM**

Al-Khwarizmi's work was written in a book titled ___

al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:293 **(Marks:1)** **Vu-Topper RM**

Random access machine or RAM is a/an

A. Machine build by Al-Khwarizmi
B. Mechanical machine
C. Electronics machine
D. Mathematical model

Question No:294 **(Marks:1)** **Vu-Topper RM**

A RAM is an idealized machine with__ random-access memory.

A. 256MB
B. 512MB
C. 100GB
D. an infinitely large

Question No:295 **(Marks:1)** **Vu-Topper RM**

What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

A. 2 n
B. 2 n n
C. n
D. 8 n

Question No:296 **(Marks:1)** **Vu-Topper RM**

Consider the following code: For(j=1; j<n;j++)
For(k=1; k<15;k++)
For(l=5; l<n; l++)
{
Do_something_constant();
}
What is the order of execution for this code.

A. O(*n*)
B. O(n^3)
C. O(n^2 log n)
D. O(n^2)

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:297 **(Marks:1)** **Vu-Topper RM**

Is it possible to sort without making comparisons?

Yes

No

Question No:298 **(Marks:1)** **Vu-Topper RM**

When we call heapify then at each level the comparison performed takes time

A. It will take Θ (1)

B. Time will vary according to the nature of input data

C. It can not be predicted

D. It will take Θ (log n)

Question No:299 **(Marks:1)** **Vu-Topper RM**

In Quick sort, we don't have the control over the sizes of recursive calls

A. True

B. False

C. Less information to decide

D. Either true or false

Question No:300 **(Marks:1)** **Vu-Topper RM**

For Chain Matrix Multiplication we cannot use divide and conquer approach because,

A. We do not know the optimum k

B. We use divide and conquer for sorting only

C. We can easily perform it in linear time

D. Size of data is not given

Question No:301 **(Marks:1)** **Vu-Topper RM**

Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. W = 50.

Item	Value	Weight
1	60	10
2	100	20

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

3	120	30

The optimal solution is to pick

A. Items 1 and 2
B. Items 1 and 3
C. Items 2 and 3
D. None of these

Question No:302 (Marks:1) Vu-Topper RM

who invented the quick sort

C.A.R. Hoare

Question No:303 (Marks:1) Vu-Topper RM

main elements to a divide-and-conquer

Divide, conquer, combine.

Question No:304 (Marks:1) Vu-Topper RM

Mergesort is a stable algorithm but not an in-place algorithm.

True
False

Question No:305 (Marks:1) Vu-Topper RM

Counting sort the numbers to be sorted are in the range 1 to k where k is small.

True
False

Question No:306 (Marks:1) Vu-Topper RM

In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

A. Convergent geometric series
B. Divergent geometric series
C. None of these

Question No:307 (Marks:1) Vu-Topper RM

In RAM model instructions are executed

One after another
Parallel

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:308 **(Marks:1)** **Vu-Topper RM**

Due to left-complete nature of binary tree, heaps can be stored in

A. Link list
B. Structure
C. Array
D. None of above

Question No:309 **(Marks:1)** **Vu-Topper RM**

The time assumed for each basic operation to execute on RAM model of computation is-----

A. Constant
B. Infinite
C. Continuous
D. Variable

Question No:310 **(Marks:1)** **Vu-Topper RM**

If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

True
False

Question No:311 **(Marks:1)** **Vu-Topper RM**

Brute-force algorithm uses no intelligence in pruning out decisions.

True
False

Question No:312 **(Marks:1)** **Vu-Topper RM**

In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

True
False

Question No:313 **(Marks:1)** **Vu-Topper RM**

For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large

True

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

False

Question No:314 (Marks:1) Vu-Topper RM
In simple brute-force algorithm, we give no thought to efficiency.
True
False

Question No:315 (Marks:1) Vu-Topper RM
The ancient Roman politicians understood an important principle of goodalgorithm design that is plan-sweep algorithm.
True
False

Question No:316 (Marks:1) Vu-Topper RM
If the indices passed to merge sort algorithm are__,then this means thatthere is only one element to sort.
A. Small
B. Large
C. Equal
D. Not Equal

Question No:317 (Marks:1) Vu-Topper RM
In pseudo code, the level of details depends on intended audience of thealgorithm.
True
False

Question No:318 (Marks:1) Vu-Topper RM
In 2d-space a point is said to be_if it is not dominated by any otherpoint in that space.
A. Member
B. Minimal
C. Maximal
D. Joint

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:319 **(Marks:1)** **Vu-Topper RM**
An algorithm is a mathematical entity that is dependent on a specific programming language.
True
False

Question No:320 **(Marks:1)** **Vu-Topper RM**
The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.
True
False

Question No:321 **(Marks:1)** **Vu-Topper RM**
F (n) and g (n) are asymptotically equivalent. This means that they have essentially the same__ for large n.
A. Results
B. Variables
C. Growth rates
D. Size

Question No:322 **(Marks:1)** **Vu-Topper RM**
8n2 + 2n - 3 will eventually exceed c2*(n) no matter how large we make c2.
True
False

Question No:323 **(Marks:1)** **Vu-Topper RM**
If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a__car.
A. Fast
B. Slow
C. Cheap
D. Expensive

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے
For More Help Contact What's app 03224021365

Question No:324 **(Marks:1)** **Vu-Topper RM**

While solving Selection problem, in Sieve technique we partition input data

A. In increasing order
B. In decreasing order
C. According to Pivot
D. Randomly

Question No:325 **(Marks:1)** **Vu-Topper RM**

In Sieve Technique, we know the item of interest.

True
False

Question No:326 **(Marks:1)** **Vu-Topper RM**

In Heap Sort algorithm, we build_for ascending sort.

Max heap
Min heap

Question No:327 **(Marks:1)** **Vu-Topper RM**

Quick sort is best from the perspective of Locality of reference.

True
False

Question No:328 **(Marks:1)** **Vu-Topper RM**

While Sorting, the ordered domain means for any two input elements x and y

_____satisfies only.

A. x < y
B. x > y
C. x = y
D. All of the above

Question No:329 **(Marks:1)** **Vu-Topper RM**

Algorithm is a mathematical entity, which is independent of a specificmachine and operating system.

True
False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:330 (Marks:1) Vu-Topper RM
In Heap Sort algorithm, the total running time for Heapify procedure is

A. Theta (log n)
B. Order (log n)
C. Omega (log n)
D. O (1)

Question No:331 (Marks:1) Vu-Topper RM
A point p in 2-dimensional space is usually given by its integer coordinate(s)___

A. p.x only p.y
B. only p.x & p.z
C. p.x & p.y

Question No:332 (Marks:1) Vu-Topper RM
In Heap Sort algorithm, the maximum levels an element can move upward is

A. Theta (log n)
B. Order (log n)
C. Omega (log n)
D. O

Question No:333 (Marks:1) Vu-Topper RM
Floor and ceiling are_to calculate while analyzing algorithms.
Very easy
Usually considered difficult.

Question No:334 (Marks:1) Vu-Topper RM
_____is one of the few problems, where provable lower bounds exist onhow fast we can sort.

A. Searching
B. Sorting
C. Both Searching & Sorting
D. Graphing

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:335 **(Marks:1)** **Vu-Topper RM**

A RAM is an idealized algorithm with takes an infinitely large random-access memory.

True

False

Question No:336 **(Marks:1)** **Vu-Topper RM**

Upper bound requires that there exist positive constants c2 and n0 such that f(n)___c2n for all n <= n0(ye question ghalat lag raha hai mujhae

A. Less than

B. Equal to or Less than

C. Equal or Greater than

D. Greater than

Question No:337 **(Marks:1)** **Vu-Topper RM**

In Heap Sort algorithm, if heap property is violated ____

A. We call Build heap procedure

B. We call Heapify procedure

C. We ignore

D. Heap property can never be violated

Question No:338 **(Marks:1)** **Vu-Topper RM**

In___we have to find rank of an element from given input.

A. Merge sort algorithm

B. Selection problem

C. Brute force technique

D. Plane Sweep algorithm

Question No:339 **(Marks:1)** **Vu-Topper RM**

A point p in 2-dimensional space is usually given by its integer coordinate(s)___

A. p.x only

B. p.y only

C. p.x & p.z

D. p.x & p.y

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:340 **(Marks:1)** **Vu-Topper RM**

In Quick Sort Constants hidden in T(n log n) are

A. Large
B. Medium
C. Small
D. Not Known

Question No:341 **(Marks:1)** **Vu-Topper RM**

The running time of quick sort depends heavily on the selection of

A. No of inputs
B. Arrangement of elements in array
C. Size o elements
D. Pivot elements

Question No:342 **(Marks:1)** **Vu-Topper RM**

Choose one Counting sort has time complexity:

A. O(n) **Page 58**
B. O(n+k)
C. O(k)
D. O(nlogn)

Question No:343 **(Marks:1)** **Vu-Topper RM**

In place stable sorting algorithm.

A. If duplicate elements remain in the same relative position after sorting
B. One array is used
C. More than one arrays are required
D. Duplicating elements not handled

Question No:344 **(Marks:1)** **Vu-Topper RM**

Cont sort is suitable to sort the elements in range 1 to k

A. K is Large
B. K is not known
C. K may be small or large
D. K is small

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:345 **(Marks:1)** **Vu-Topper RM**

The sieve technique works where we have to find_item(s) from a large input.

A. Single
B. Two
C. Three
D. Similar

Question No:346 **(Marks:1)** **Vu-Topper RM**

In strong components algorithm, first of all DFS is run for computing finish times of vertices.

A. [d[u], f[u]]⊆[d[v], f[v]]
B. [d[u], f[u]]⊇[d[v], f[v]]
C. unrelated
D. Disjoint

Question No:347 **(Marks:1)** **Vu-Topper RM**

For the sieve technique we solve the problem,

A. recursively
B. mathematically
C. precisely
D. accurately

Question No:348 **(Marks:1)** **Vu-Topper RM**

number of nodes in a complete binary tree of height h is

A. 2^(h+1) – 1
B. 2 * (h+1) – 1
C. 2 * (h+1)
D. ((h+1) ^ 2) – 1 25

Question No:349 **(Marks:1)** **Vu-Topper RM**

Heaps can be stored in arrays without using any pointers; this is due to the

_____nature of the binary tree,

A. left-complete
B. right-complete
C. tree nodes

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

D. tree leaves

Question No:350 (Marks:1) Vu-Topper RM

Sorting is one of the few problems where provablebonds exits onhow fast we can sort,

A. upper
B. lower
C. average
D. log n

Question No:351 (Marks:1) Vu-Topper RM

The analysis of Selection algorithm shows the total running time is indeed
_____in no

A. arithmetic
B. geometric
C. linear
D. orthogonal

Question No:352 (Marks:1) Vu-Topper RM

What algorithm technique is used in the implementation of Kruskal solution forthe MST?

A. Greedy Technique
B. Divide-and-Conquer Technique
C. Dynamic Programming Technique
D. The algorithm combines more than one of the above techniques

Question No:353 (Marks:1) Vu-Topper RM

one If you find yourself in maze the better traversal approach will be :

A. BFS
B. BFS and DFS both are valid
C. Level order
D. DFS

Question No:354 (Marks:1) Vu-Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Suppose that a graph G = (V,E) is implemented using adjacency lists. What isthe complexity of a breadth-first traversal of G?

A. O(|V |^2)
B. O(|V | |E|)
C. O(|V |^2|E|)
D. O(|V | + |E|)

Question No:355 (Marks:1) Vu-Topper RM

There are_nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

A. 2
B. 3
C. 4
D. 5

Question No:356 (Marks:1) Vu-Topper RM

Counting Money problem is an example which cannot be optimally solved bygreedy algorithm.

True
False

Question No:357 (Marks:1) Vu-Topper RM

Who invented Quick sort procedure?

A. Hoare
B. Sedgewick
C. Mellroy
D. Coreman

Question No:358 (Marks:1) Vu-Topper RM

An activity scheduling algorithm, the width of a rectangle --.

A. Is always ignored
B. Direct toward recursion
C. Should be maximized
D. Indicates the duration of an activity

Question No:359 (Marks:1) Vu-Topper RM

In recursive formulationof Knapsack Problem: V[0, j] = for

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

j>=0

A. -1

B. 0

C. 1

D. 2

Question No:360 (Marks:1) Vu-Topper RM

The Probability in Huffman encoding is the number of occurrences of acharacter divided by the total----------in the message.

A. Numbers

B. Frequencies

C. Strings

D. Characters

Question No:361 (Marks:1) Vu-Topper RM

Graphs can be represented by an---------- and --.

A. queue , stack

B.adjacency list , adjacency matrix

C.adjacency right , adjacency left

D.Binary , linear

Question No:362 (Marks:1) Vu-Topper RM

According to parenthesis lemma vertex u is a descendent of v vertex if and only if:

f [d[u], f[u]] ⊆ [d[v], f[v]] (Page 129)

Vu Topper RM

Visit My YouTube Channel For Subjective and More Important Files

For More Help Contact What's app 03224021365

Channel Name = #VuTopperRM

VuTopperRM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365